Zoo 404

Physiology

Lecture: 01

Spectrum of study of physiology

*. **Physiology:**

def:

"Study of functions of tissues, organs, and organ system of animals."

- To understand mechanism operate in animals.

* Physiological obeys laws of physics and chemistry:

- Ohm's law
- Boyle's law
- Ideal gas law
- law of gravity
- Kinetic and Potential energy.
- Inertia
- momentum
- velocity
- Drag

⇒ Curiosity underlies learning of physiology.

How a humming birds heart beats 20 times per sec during hovering flight??

How can insect see in ultraviolet Spectrum???

How kangaroo rats survive in deserts???

*- Animal Physiology to human physi- :

•- An insight into physiological Processes of humans.

•- human Species have linked evolutionary history.

e-g

human heartbeat, elective nerve impulse

*- Animal physiology lays foundation of human physiology.

*- human physiology lays foundation of Scientific medical practice.

*= Developing treatment of human diseas.

Imp models for human diseases

•- Diabetic mice fat cats.

•- zebra fish embryos with heart defects.

lect#02

Principle of Homeostasis

•- Walter Cannon (1929) MCQ's

Def: "Tendency of organisms to maitain relative internal stability, The ability to Protect internal fluctuations harms of fluctuation in external environment called homeostatic"

* Environmental fluctuation:

most habits quite hostile to animal cell.

eg: freshwater more dilute

Seawater more salty.

Enviromental temp.. may be to hot or too cold.

* Need for homostatic mechanism:

Environmental fluctuation disruptive to funtions of cells, tissues, organs.

Physiology regulatory maintain Stable conditions

Homestatic : eg

Regulation of water, Regulation of temp..

Reg of Ph, Reg of glucose Conc, Reg of osmotic Ptessure

Reg of oxygen level, Reg of ion Conc...

mechanism:

•-Variables •- Set point •- Living Control System vs physical control System

Three Components: 1- Receptors

2- Control centers effectors.

eg

Temp.. Control System in air Conditionats and water heating gysers vs Endothermic animals.

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Lecture: 03

Feedback Control System

Def:

mechanism of self regulations of biological processess.

Principle:

output or product itself Regulates the process.

Significance:

maintenance of homstasis.

feedbact controls Regulate particular variables e.g, Temperature, Salinity, ph.

Regulation Requires:

Continues Sampling of Controlled variables

Corrective actions.

Types:

- Negative feedbact System
- Positive Feedback System

Negative feedback System:

most Common form of regulation.

end product works to stop or slow down process.

E.g:

- Breakdown of sugar in cell generates ATP

Control of Blood sugar (glucose)

Positive feedback:

end product speeds up process of its production.

e.g:

clotting of Blood to response an injury

Release of oxytocin during Labor.

Lecture # 04

Conformity and Regulation

Two catagories of animals Responses to change in external environment-

* Conformity:
* Regulation.

Conformers:

Animal: that conform to external change unable to maintain homeostasis for internal Conditions.

Types:

- Osmoconformers
- Oxyconformers
- ThermoConfetmers.

Regulator:

Use internal control mechanisms to regulate internal conditions.

* maintain homeostasis.

- osmoregulators
- oxyregulators
- Thermoregulators

Lecture #05

"Adaptation"

'A slowly occuring evolutionary process.'

- Involves thousands of generations
- Physiology of Species becomes well matched to environment.
- Ensure Survival of species.
- Not reversible.

- Example:
  - Bar headed goose
    - Able to fly above mount Everest.
  - Species adapted to high altitude.

Acclimatization:

'An Adaptive change within an individual animal'

- Happens due to chronic, exposure to new, naturally occuring environmental conditions.

Acclimation:

'Changes are induced experimentally in Labortary'

⇒ Both of these are:

- Acquired characters, • Restricted to one or few members of species
- not inheritable, • no evolutionary
- Reversible.

e.g:

An Animal migrates to high altitude area.
faces low oxygen partial pressure.

* effects:

- Lung ventillation rate increase.
ventillation drop to nomal in few days.

Lecture: 06

"membrane Permeability"

* Selective Permeability:

Structural features of plasma membrene confer it with selective permeability?

- PM regulate movement of substance across it.
maintenance of intercellular physiological conditions.
- Fundamental to functioning of living cells and maintenance of intercellular physiological Conditions.

membran permeability for various substance:

Varies greatly with type and size of molecules.

non-polar, lipid soluble substance:

Cross passively by ~~passing~~ dissolving in lipid bilayer of membrane.

Rate of Pentration depends on Lipid Solubility and size of molecules

Polar Substance & ions:

have difficulty in passing through because of hydrophobic interior of lipid Layer.

macromolecules Can't pass membrane due to size.

***- Passive permeability and Permeability Constant:**

Permeability of membrane is the rate at which Substance Passively pentrates the membrane.

$$\frac{dQ_s}{dt} = P(C_I - C_{II})^3$$

C_1, C_2 (Concentrations).
P (permeability constant).

$\frac{dQ_s}{dt}$ amount of substance...

imp • - Permeability can be altered.
e.g: Antidiuretic hormone can be increase water permeability of renal Collecting duds in mammals upto 10mints

Lecture# 07

imp
"molecules have thermal energy due to which they are in Continuous motion Called thermal motion.

• - One result of T-motion is diffusion.
"Random thermal motion of molecules result in their dispersion from region of higher Concentration to lower Conc.. Called diffusion."
e.g diffusion of Oxygen.

Diffusion is passive transport.
Occur without expense of energy.
Concentration act as potential energy.

mcq: Rate of diffusion varies from different molecules to different Permeability.

Rate of diffusion of solute Calculate by fick diffusion equation.

$$\frac{dQ_s}{dt} = D_s A \frac{dc_s}{dx}$$

$\frac{dQ_s}{dt}$, rate of diffusion. D_s diffusion Coefficient of S. Varies with nature and molecular weight of substance and solvent.
A, Cross Sectional Area which S is diffusing.
$\frac{dc_s}{dx}$ is Concentration gradient, with distance dx. determine rate which S will diffuse.

"The amount of Solute passes through a unit area of membrane every Second in One direction called membrane flux"

$$J = \frac{dQ_s}{dt}$$

flux.

- if Solute Occure both sides of membrane
- it exhibits flux in each direction flux in one direction independent flux in opposite direction

mcQsimp - if influx and eflux is equal then net flux is zero.
- if flux is greater in one direction net flux is different between two fluxes.

Lecture # 8

"Passive diffusion through membrane down Conc. gradient with help of transport proteins called faciliated diffusion"
- It is passive process, not requiring energy in form of ATP.

Transport proteins very Specific.
Transport Some Substance not others.
- Types of transport Proteins:

*. Channel proteins *. Carrior proteins
- Channel Proteins:

Extend across cell membrane.
Have filtered pores membrane channels.

membrane channels: Allow Specific molecules or ion to pass through and cross membrane. Diameter of channels is less th

Can opening and close and is regulated. molecules can cross membrane when pores open.

Types

- water channel proteins
- Ion channel protein
- Specialized channel proteins.

* **water channel proteins:**
- Known as Aquaporins.
- facilitate diffusion of water and Small ions e.g Na^+, K^+, Cl^-
- Have hydrophilic passageways that facilitate quick flow of water molecules and ions in dissolved form.

* **Ion channel proteins:**
- Allow Specific ions.
- Gated channels.
- open or close response to electrical or chemical stimulus.
- chemical stimulus is Substance other than is to be transported.

e.g neurotransmitters for Na^+ gated ion channels.

- Electrical stimulus activate K^+ ion channels.

*-Specialized channels proteins:

- many channels specialized for faciliated diffusion of small, uncharged polar molecules. e.g CO_2, Co, NO.

Carrier Proteins:

- They are very selective.
- Movement of polar molecules e.g Sugar amino acid nucleotides.

Types

- Exist in many forms in all types of membranes.

⇒ various functional types:

- Uniportes
- Couple transporters.
- Symptors
- Antiporters.

Lecture # 09:

Donnan Equilibrium

Discovery:

The phenomenon of unequal distribution of ions across two Sides of differentially permeable membrane.

Discovered by fredrick Donnan in 1911 Given the Donnan equilibrium.

"if diffusible Solutes are Separated by membrane that freely permeable to water and electrolytes but totally impermeable to one species of ions. diffusible Solutes unequally distributed b/w two compartments"

Explaination:

Doman Equilibrium charactarized by ~~catoi~~ reciprocal distribution of anions and cations.

gradient not only chemical but electrical.

electronegativity with both compartments positive charges and negative must equal.

Doman effect:

Concentration of Solution with non-diffusible Solute remains high at equilibrium. This is donnan effect.

Lecture # 10

Ion Distribution Along Across Plasma Membrane

plasma membrane maintain different Conc. of ions inside and outside cell. major ions that unequally distributed are k^+, Na^+, Cl^-, Ca^{2+}.

Distribution of k^+, Na^+, Cl^-, Ca^{2+}

- The Conc. of k^+ maitained 10–30 times more than Cytosol than extracellular fluid. because cell membranes more permeable to k^+ and k^+ more Conc inorganic ions in cytosol.
- Intracellular conc. of Na^+ and Cl^- ions in maitained about ten time lower than extracellular.
- Cytosolic Conc. of Ca^{2+} remains below 10^{-6} M.